

Théo Alessandro Hermann

Researcher & Perception Engineer · MIT Senseable City Lab Rio

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MSc Robotics, EPFL · 4 years shipping perception and autonomy systems on edge hardware · Franco-Brazilian, trained in Switzerland.

Founding member of **MIT Senseable City Lab Rio** — researcher on the 3D digital twins of favelas line, reconstructing dense informal settlements from LiDAR for public health work. On the side, building open-source robotics software. Taking a small number of consulting engagements in **perception and AI-agent infrastructure**.

Previously at **Roboat** (MIT spinoff) — built the perception stack for autonomous canal vessels deployed in Amsterdam.

EXPERIENCE

Research Fellow — Founding Member, Rio branch Jan 2025 – Present
MIT Senseable City Lab Rio · Rio de Janeiro, Brazil

Researcher on the urban-digital-twin line — high-fidelity 3D reconstructions of favelas from terrestrial LiDAR and aerial imagery, feeding OpenFOAM CFD simulations of airborne disease transmission. Tech: Python, Blender, GeoPandas, Google Earth Engine.

Perception Engineer — First Engineering Hire Mar 2024 – Aug 2024
Roboat (MIT Spinoff) · Amsterdam, Netherlands

Designed and shipped end-to-end perception stack for autonomous canal vessels. YOLO-based detection + multi-object tracking at <30ms on Jetson Orin. Built full MLOps pipeline with DVC over 100+ hours of maritime data. LiDAR point cloud segmentation for dynamic water filtering. Tech: PyTorch, ROS2, Docker, DVC.

Visiting Research Fellow Oct 2023 – Feb 2024
IRI (CSIC-UPC) · Barcelona, Spain

Built PyTorch fusion model combining neuromorphic event cameras with visual-inertial odometry for 6-DOF drone state estimation during aggressive maneuvers (>5g). Tech: PyTorch, ROS, C++.

Research Assistant Jun 2022 – Jun 2023
VITA Lab, EPFL · Lausanne, Switzerland

Built and open-sourced a full ROS autonomy stack for mobile robot navigation in crowded environments. Real-time detection, multi-object tracking, 3D pose estimation, and trajectory prediction on Segway Loomo. Tech: ROS, Python, C++, OpenCV.

EDUCATION

Master’s degree (M.Sc), Robotics — EPFL 2021–2024

Exchange (B.Sc 3rd year), Embedded Systems — KTH Royal Institute of Technology 2020–2021

Bachelor’s degree (B.Sc), Microengineering — EPFL 2018–2021

TECHNICAL SKILLS

Robotics	ROS/ROS2, SLAM, path planning, sensor fusion, Jetson/edge deployment
ML / AI	PyTorch, scikit-learn, LLM integration, RAG systems, LlamaIndex, reinforcement learning
Perception	LiDAR processing, point clouds, stereo vision, event cameras, 3D reconstruction
Dev	Python, C++, Git, Docker, CI/CD, DVC, MLOps
Tools	OpenFOAM CFD, Blender, MATLAB, GeoPandas, QGIS

LANGUAGES

French Native · **English** Fluent · **Portuguese** Fluent · **Spanish** Conversational